

EDUCATION

Carnegie Mellon University <i>M.S. in Software Engineering</i> GPA: 4.0 / 4.0 Selected Coursework: Large Language Model Systems, Machine Learning, Distributed Systems	Expected May 2027
University of Wisconsin–Madison <i>B.S. in Computer Science & B.S. in Computer Engineering</i> GPA: 3.72 / 4.0 Selected Coursework: Large Language Models in Practice, Operating Systems, Computer Architecture	May 2024

WORK EXPERIENCE

NewsBreak <i>AI Engineer Intern</i>	May – August 2026 <i>Mountain View, CA</i>
- Designed and shipped a personalized LLM alert pipeline as a Kubernetes CronJob (bge embeddings → LLM scoring → per-topic percentile gate → LLM dedupe → push, ~1,200 LLM calls/day). - Evaluated it over 11 days of production shadow (4,100+ logged decisions) and cut filler 34% → 4%. Hand-graded alerts showed non-events scored like real ones (median 89 vs. 90), so I added an event gate. - Rejected a rule that caught 91% of filler but killed the day’s two highest-reach real alerts, and retired my own first detector after 19 hours of production data showed it was 15 hours too slow (net –882 lines). - Shipped agentic search over SSE (0.85s to first token): deterministic RAG on turn 1, a bounded tool-calling agent (≤3 hops, 5 tools) on follow-ups, hybrid retrieval fused by RRF, and server-resolved citations. Entity resolution took real logged queries from 0% to 100% P@1 at 0.4ms. - Root-caused a 65% failure rate in another team’s LLM workers via Langfuse traces as an unbounded reasoning loop (p95 1,197 tokens vs. 16,384 runaways) and a Helm env shadowing the code default. - Stood up the service on AWS EKS (Docker, Helm, Jenkins) on day 3, set up Prometheus and Grafana, and fixed a 5-day silent CronJob outage (non-terminal ImagePullBackOff + Forbid concurrency). - Built the Python FastAPI backend from its first commit to 89 endpoints with ~1,450 offline tests and cut a 25–30 request fan-out to one call on a shared cache 78× smaller than per-user keys. - Sole SwiftUI engineer and primary backend author for SportsBreak (1.4K App Store ratings, 1,119 peak daily active users), and wrote 34 contract-first API specs for the team.	
Haddee Education <i>Engineering Lead</i>	June – September 2025 <i>San Diego, CA</i>
Led an 8-intern team through Agile sprints to ship a React, Next.js, FastAPI, and Supabase web app from 0→1 on GCP App Engine and designed the database schemas and REST APIs for its LangChain LLM features.	
ZenAI — MiraclePlus (Formerly YC China) 2023 Fall Batch <i>Full Stack Developer</i>	September – December 2023 <i>Beijing, China</i>
Built a VC-backed startup’s LLM mock-interview web app from 0→1 as the sole engineer, then demoed it and iterated with clients until three 200–500-person companies signed contracts.	

RESEARCH EXPERIENCE

Control & Learning Group — Carnegie Mellon University <i>Graduate Researcher</i> Advisor: Prof. Yorie Nakahira	February – June 2026
Mapped natural-language instructions to formal safety specs with multi-turn LLMs and implemented Bayesian estimators for real-time road-friction adaptation (LuGre tire models).	

SKILLS

Languages	Python, Go, Java, SQL, Bash, TypeScript, Swift, C
AI/ML	PyTorch, CUDA, RAG, hybrid retrieval, bge embeddings, LLM tool calling / agents, Langfuse
Infra	Docker, Kubernetes (EKS), Helm, AWS, GCP Vertex AI, Prometheus, Grafana, CI/CD, Linux
Backend	FastAPI, Pydantic, asyncio, Server-Sent Events (SSE), MongoDB, PostgreSQL, pytest
Certifications	AWS Certified Solutions Architect – Associate, AWS Certified Cloud Practitioner

PROJECTS

- codingplans:** Building a site that rates LLM coding plans. Tested Z.AI, MiniMax, and Kimi on KV caching, long context, and needle-in-a-haystack to show which nerfed caching or heavily quantized models.